

PREDICTING THE INFECTION OF COVID-19 IN PATIENTS FROM SARS-CoV-2 CT SCAN DATASET USING DIFFERENT CNN MODELS

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Abstract

This is a project that tries to predict whether a patient is infected by a disease called Covid-19 or not. This disease was spread by SARS-CoV-2, during the 2019-2022 period when WHO declared it as a pandemic due to its huge and rapid widespread infecting, also killing many people. They are caused by kind of infection in the Thorax part of the body that is the chest. The virus moves down to the lungs, causing pneumonia, which reduces oxygen saturation (hypoxia). The severe inflammatory response can lead to systemic failures beyond the lungs, including heart failure, septic shock, and damage to other organs, ultimately leading to death. The symptoms start appearing 1-14 days after first exposure. The ways of testing a patient include Molecular Testing (RT-PCR), Antigen Testing, Imaging (Chest CT Scan), Serology (Antibody Test). A CT scan is superior because it provides a direct visual map of physical lung damage and inflammation that molecular tests (like PCR) and other tests cannot see prominently.

Keywords

CT scan, CNN, deep learning, Transfer Learning, COVID-19

Introduction

Coronavirus disease (COVID-19) is an infectious illness caused by the SARS-CoV-2, primarily affecting the respiratory system. It emerged in late 2019 and spread globally, leading the World Health Organization to declare it a pandemic in 2020. The topic of this paper of finding the accuracy of Covid-19 prediction from CT Scan was done by many researcher in the near past, providing me the insights to write this paper. The process is that we need to collect CT Scan of the patient and test it if predicts it correctly as per the training data set feeded to the CNN(Convolution Neural Network) which could help us to identify and treat the patient back then faster.

Paper 1: Diagnosis of COVID-19 using CT scan images and deep learning techniques

This paper presented **by Shah, Vruddhi, et al.** provides an insightful on the topic, they have created their own model named "CTnet-10" and used other baseline models(DenseNet-169, VGG-16, ResNet-50, InceptionV3, and VGG-19) to compare the different accuracy of the model. They collected their data from open-source dataset called "COVID-CT dataset", originally built by Zhao et al. Images collected from four hospitals/ institutions in Hunan, China consisting of 738 chest (thorax) CT images, the dataset was divided into training (80%), validation (10%), and testing (10%) sets. The VGG-19 proved to be superior with an accuracy of 94.52% as compared to all other deep learning models and their self-developed model, Ctnet-10 shows an accuracy of 82.1%. The study demonstrates that thorax CT scan-based deep learning models can be an efficient and rapid method for early COVID-19 detection, supporting clinical diagnosis.

More on this in here-: [Diagnosis of COVID-19 using CT scan images and deep learning](#)

Paper 2: Deep transfer learning-based automated detection of COVID-19 of lung CT scan slices

This paper **by Ahuja et al.** presents a method for detecting COVID-19 using chest (thorax) CT scan images and deep learning techniques. The authors proposed a three-phase model consisting of data augmentation, disease detection, and abnormality localization to improve the performance of COVID-19 classification. The dataset includes 349 COVID-19 positive images from 216 patients and around 397 non-COVID images. Due to the limited dataset, the authors applied data augmentation techniques to avoid overfitting. The models used include ResNet18, ResNet50, ResNet101, and SqueezeNet, which classify CT scan images into COVID-19 and non-COVID categories. Among the models, the best performance was achieved using ResNet18 pre-trained transfer learning-based model on testing accuracy up to 99.4. The study concludes that combining transfer learning with data augmentation can effectively improve COVID-19 detection accuracy from CT scan images, even when the dataset size is limited. It also helps in identifying abnormal regions in the lungs, making it useful for clinical diagnosis.

More on this paper -: [Deep transfer learning-based automated detection of COVID-19 of lung CT scan slices](#)

Implementation of Custom Architectures vs. Pre-trained Models: We have tried to use popular architectures like VGG16, VGG19, and ResNet50 for transfer learning, but you've also built custom versions of CNNs (CNN1, CNN2) and even re-implemented the ResNet50 architecture from scratch.

Comparative Analysis of Different Architectural Approaches: The systematically compared a simple CNN, a CNN with Batch Normalization, custom VGG/ResNet implementations, and transfer learning approaches (VGG16-TL, VGG19-TL, ResNet50-TL).

Investigation of Batch Normalization's Impact: The CNN2 explicitly adds Batch Normalization to CNN1. This allows for a direct comparison of the effect of Batch Normalization on training stability, speed, and final accuracy.

Methodology

The dataset used is a public available SARS-CoV-2 CT scan dataset, containing 1252 CT scans that are positive for SARS-CoV-2 infection (COVID-19) and 1230 CT scans for patients non-infected by SARS-CoV-2, 2482 CT scans in total. These data have been collected from real patients in hospitals from Sao Paulo, Brazil.

It has two separation as COVID(1,252 images) and NOT-COVID(1,230 images) which is quite a raw data so for using it in my project I used to first create two files using program of train and test files and they are split in the ratio as **80% Train / 20% Test** using **python libraries like os,shutil and random**.

Then we did data augmentation and preprocessing on the data using **ImageDataGenerator from the tensorflow.keras.preprocessing.images package** so that in every epoch the model gets new picture instead of old one also it does not make the hard drive more bulky with new images instead uses ram for augmentation.

We have used in total 8 models to understand the performance on the data and the accuracy of the models changing for different models

General CNN model

A Convolutional Neural Network (CNN) is a specialized type of Deep Learning architecture designed to process data with a grid-like topology, most commonly images. Unlike standard neural networks that treat every pixel as an independent variable, a CNN uses spatial hierarchy. It identifies simple patterns (edges, textures) in the initial layers and combines them into complex concepts (shapes, objects) in the deeper layers.

Our Specialized model **CNN1** follows a sequential pipeline. It has 5 convolution Layers having Conv2D of 3*3 size kernel and 256 in number, uses ReLu activation function and MaxPooling of 2*2 and 5 in number. It has a 2 Fully Connected Layer of 128 and 64 perceptrons units each also dropout of 0.2 is given to stop overfitting. Classification is done using softmax and Adam optimiser.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss: 0.6932 , Train Accuracy: 0.5045

After which we evaluated the data using the test_generator dataset and got result as:-

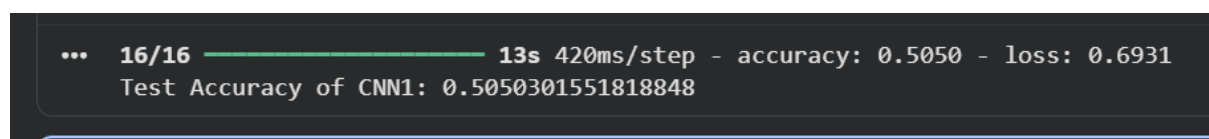


Fig:-The image shows the test accuracy and loss of the Simple CNN models

Batch Normalised CNN

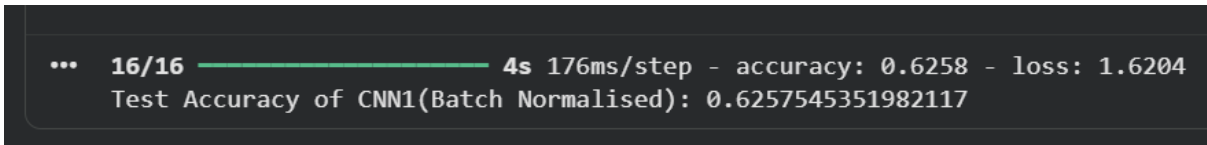
Batch Normalization (BN) is a technical optimization that standardizes the inputs to each layer within a deep network. By maintaining a consistent mean and variance for layer activations, BN accelerates the training process, allows for higher learning rates, and provides a regularization effect that stabilizes the model's overall learning trajectory.

This enhanced model **CNN2** follows a sequential pipeline integrated with normalization layers. It features 5 Convolutional Blocks, each consisting of a Conv2D layer with 3 * 3 size kernels, a Batch Normalization layer, and 2 * 2 MaxPooling. The architecture utilizes the ReLU activation function throughout to ensure non-linearity. It has 2 Fully Connected Layers of 128 and 64 units respectively, both accompanied by Batch Normalization to mitigate internal covariate shift. A Dropout of 0.2 is applied to prevent overfitting, while the final classification is performed using Softmax and the Adam optimizer.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss: 0.8765, Train Accuracy: 0.2846

After which we evaluated the data using the test_generator dataset and got result as:-



```
... 16/16 4s 176ms/step - accuracy: 0.6258 - loss: 1.6204
Test Accuracy of CNN1(Batch Normalised): 0.6257545351982117
```

Fig:-The image shows the test accuracy and loss of the Batch Normalised CNN models

Implementing VGG-16 from scratch

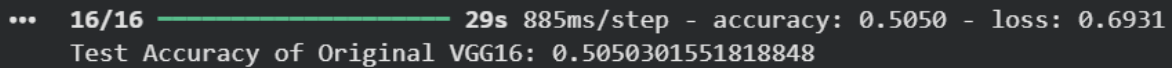
VGG16 (Visual Geometry Group) is a renowned deep convolutional neural network architecture characterized by its depth and the consistent use of very small 3*3 convolution filters. By stacking multiple convolutional layers before pooling, the model increases its receptive field and captures highly complex features.

This implementation of VGG16 follows a sequential pipeline divided into 5 Convolutional Blocks. The first two blocks contain 2 convolutional layers each, while the subsequent three blocks feature 3 convolutional layers each, totaling 13 Conv2D layers using 3 * 3 kernels and ReLU activation. Every block concludes with a 2*2 MaxPooling layer with a stride of 2. The classification head consists of 2 Dense layers with 4096 units each for high-level feature reasoning, followed by a final Softmax output layer. The model is compiled using the Adam optimizer and categorical crossentropy.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss: 0.5045 , Train Accuracy: 0.6932

After which we evaluated the data using the test_generator dataset and got result as:-



```
... 16/16 ————— 29s 885ms/step - accuracy: 0.5050 - loss: 0.6931
Test Accuracy of Original VGG16: 0.5050301551818848
```

Fig-:The image shows the test accuracy and loss of the VGG16 models

Implementing VGG-19 from scratch

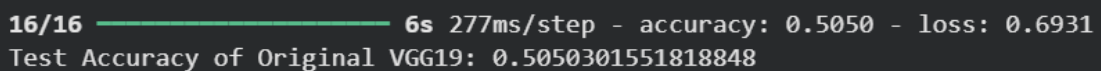
VGG19 is an advanced iteration of the Visual Geometry Group architecture, further increasing network depth to improve feature representation. By utilizing 19 weight layers, the model can capture more granular details and complex hierarchical patterns.

This implementation of VGG19 utilizes a sequential pipeline consisting of 16 Convolutional layers organized into 5 distinct blocks. The first two blocks contain 2 convolutional layers each, while the following three blocks feature 4 convolutional layers each, all employing 3*3 filters with "same" padding and ReLU activation. Each block is partitioned by a 2*2 MaxPooling layer to downsample spatial dimensions. The network concludes with a high-capacity classification head featuring 2 Dense layers of 4096 units and a final Softmax output layer for multi-class assignment, optimized via the Adam algorithm.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss: 0.5214 , Train Accuracy: 0.6928

After which we evaluated the data using the test_generator dataset and got result as:-



```
16/16 ————— 6s 277ms/step - accuracy: 0.5050 - loss: 0.6931
Test Accuracy of Original VGG19: 0.5050301551818848
```

Fig-:The image shows the test accuracy and loss of the VGG19 models

Implementing ResNet50 from scratch

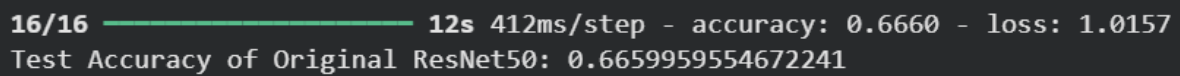
ResNet50 (Residual Network) is a deep learning architecture that introduced the concept of Residual Learning to solve the vanishing gradient problem in very deep networks. By utilizing "skip connections" or shortcuts, the model allows gradients to flow through the network more effectively.

This model ResNet50 implementation is structured into five distinct stages using a functional API approach. The architecture begins with a Lambda layer to convert grayscale input into three-channel RGB, followed by an initial 7*7 convolution and Max-Pooling. The core of the model consists of multiple Identity and Convolutional blocks, which utilize a bottleneck design (1x1, 3x3, and 1x1 convolutions) to manage computational complexity. In total, the model transitions through stages of increasing filter depth—from 256 to 2048 filters. The process concludes with Global Average Pooling and a Softmax activation layer for classification, optimized via the Adam algorithm and categorical crossentropy.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss: 0.9140 , Train Accuracy: 0.2358

After which we evaluated the data using the test_generator dataset and got result as:-



```
16/16 ————— 12s 412ms/step - accuracy: 0.6660 - loss: 1.0157
Test Accuracy of Original ResNet50: 0.6659959554672241
```

Fig:-The image shows the test accuracy and loss of the ResNet50 models

Transfer Learning with VGG16

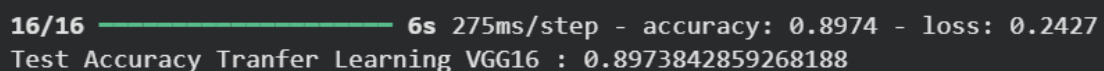
Transfer Learning via the VGG16 architecture utilizes pre-trained weights from the ImageNet database to accelerate the convergence of deep networks on specialized datasets. By repurposing low-level feature extractors—such as edge and texture detectors—the model maintains high predictive performance.

This Fine-tuned VGG16 model implements a specialized pipeline starting with a Concatenate layer that replicates the single-channel grayscale input across three dimensions to achieve RGB compatibility. The core architecture utilizes the VGG16 frozen base to extract complex features without altering early-stage weights. A robust classification head is appended, featuring Flattening, two Dense layers (256 and 128 units), and high-ratio Dropout (0.5) to prevent overfitting. To optimize the model for the target classes, Fine-tuning is applied by unfreezing the final four layers of the base model, allowing for subtle weight adjustments via the Adam optimizer at a highly conservative learning rate of $1 * 10^5$.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss:0.8730 , Train Accuracy: 0.3181

After which we evaluated the data using the test_generator dataset and got result as:-



```
16/16 ————— 6s 275ms/step - accuracy: 0.8974 - loss: 0.2427
Test Accuracy Tranfer Learning VGG16 : 0.8973842859268188
```

Fig:-The image shows the test accuracy and loss of the Transfer Learning VGG16 models

Transfer Learning with VGG19

Transfer Learning is a paradigm where a model developed for one task is reused as the starting point for a model on a second task. It utilizes the pre-trained weights of VGG19—originally trained on the massive ImageNet dataset—the model leverages pre-existing knowledge of low-level visual features.

This Fine-tuned VGG19 model employs a hybrid strategy. First, a Concatenate layer is used to transform the single-channel grayscale input into a three-channel RGB format to maintain compatibility with the pre-trained architecture. The base features are extracted using the VGG19 frozen weights, ensuring that general visual knowledge is preserved. A custom classification head is then appended, consisting of Flattening, two Dense layers (256 and 128

units), and Dropout (0.5) for heavy regularization. To optimize performance for our specific dataset , Fine-tuning is performed by unfreezing the final four layers of the VGG19 base, allowing the model to adapt its high-level filters while using a minimized learning rate of $1 * 10^5$ via the Adam optimizer.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss:0.8710 , Train Accuracy: 0.3241

After which we evaluated the data using the test_generator dataset and got result as-:

```
16/16 ————— 6s 318ms/step - accuracy: 0.9054 - loss: 0.2465
Test Accuracy of Tranfer Learning VGG19: 0.9054325819015503
```

Fig:-The image shows the test accuracy and loss of the Transfer Learning VGG19 models

Transfer Learning with ResNet50

Transfer Learning via the ResNet50 architecture utilizes "Deep Residual Learning" to mitigate the degradation problem associated with increasing network depth. By incorporating pre-trained ImageNet weights, the model uses optimized spatial filters .

This Fine-tuned ResNet50 model integrates a Concatenation layer at the input stage to replicate the single-channel grayscale data into a three-channel format, ensuring architectural compatibility. The model maintains a frozen base for the majority of its 50 layers to preserve general feature extraction capabilities. To customize the network for our specific dataset, we implement a Fine-tuning strategy by unfreezing the final 10 layers of the ResNet base. The classification head is updated with a Flattening layer, a 256-unit Dense layer with ReLU activation, and a 50% Dropout for robust regularization. The final Softmax layer is optimized using a controlled learning rate of $1 * 10^4$ with the Adam optimizer.

Then we train the model with the train_generator dataset for 20 epochs giving a

Train Loss: 0.7087 , Train Accuracy: 0.5777

After which we evaluated the data using the test_generator dataset and got result as-:

```
16/16 ————— 10s 306ms/step - accuracy: 0.7706 - loss: 0.4765
Test Accuracy of Transfer Learning ResNet50: 0.7706237435340881
```

Fig:-The image shows the test accuracy and loss of the Transfer Learning ResNet50 models

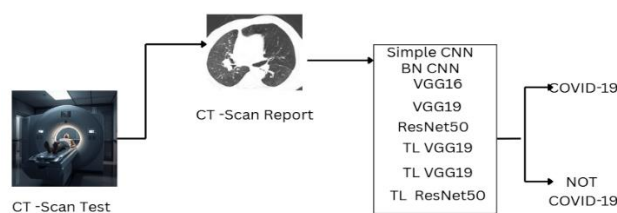


Fig:-The figure shows the entire process of using the models for prediction

Result and Analysis

Below is shown the confusion matrix of the models to help us distinguish COVID-19 and NOT-COVID-19 predicted by different models showing the accuracy of the models in the order:-

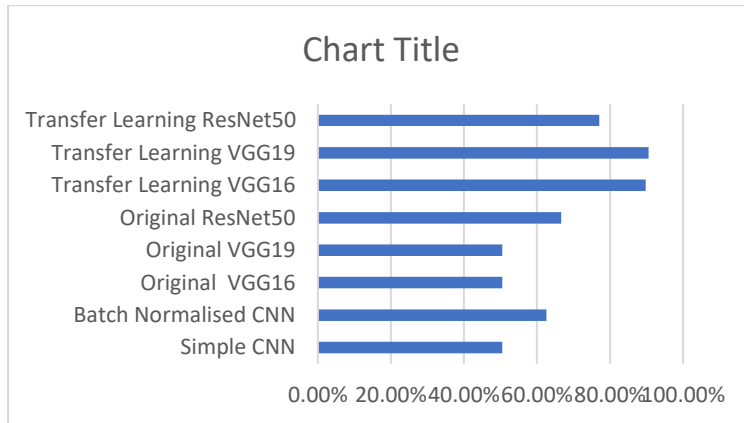


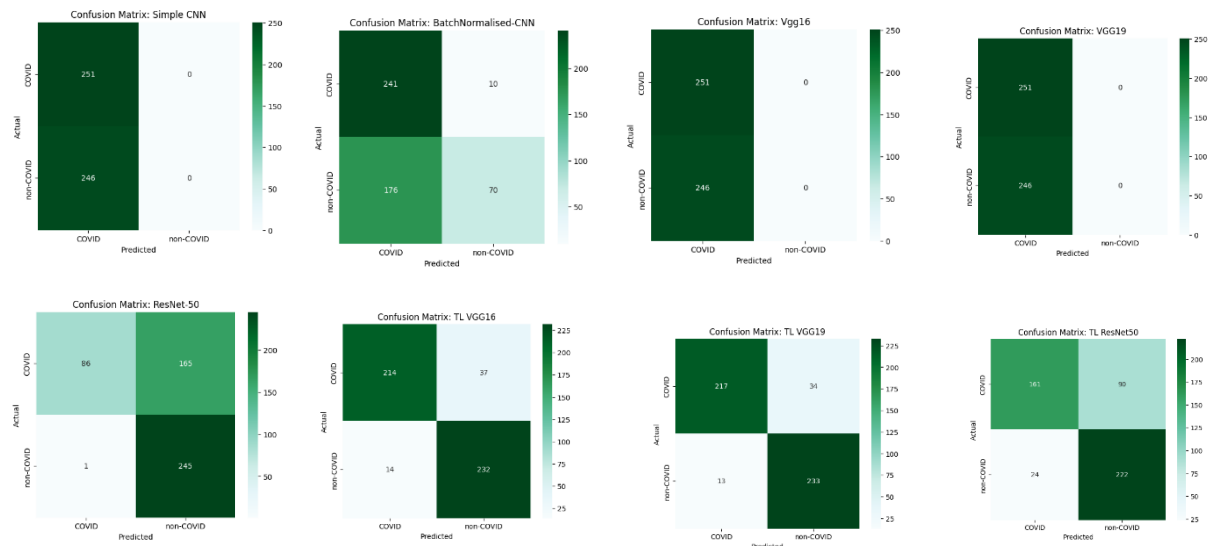
Fig:-The bar chart shows the different test accuracy of the models

Tranfer Learning Vgg19 > Tranfer Learning Vgg16 > Tranfer Learning ResNet50 > Original Resnet50 > Batch Normalised CNN > Simple CNN > Original VGG16 > Original Vgg19

Fig:-The table shows the different test characteristics of the models

Model	Accuracy Percentage	Total Layers	Key Feature
CNN1	50.5%	5 Conv	Baseline Architecture
CNN2	62.57%	5 Conv	Batch Normalization
VGG16	50.5%	13 Conv	Uniform 3*3 Kernels
VGG19	50.5%	16 Conv	Increased Depth
ResNet50	66.59%	50 Layers	Skip Connections
Tranfer VGG16	89.73%	13 Conv	RGB Concatenation
Tranfer VGG19	90.54%	16 Conv	RGB Concatenation
Transfer ResNet50	77.06%	50 (Residual)	Residual Connections

Fig:-Shows the different Confusion Matrix of the models to understand their performance



Drawbacks

1. This dataset has a range of dimension of the image but we did it with a particular dimension that is 256×256 as the base which effected the result
2. The dataset is greyscale but model like ResNet50 specially can work with high efficiency on RGB images (we passed it through 3 channels ,that compromised the training as well as testing)
3. Some models like Simple CNN, Original implementation of VGG16, Original implementation of VGG19 did not performed well as it cannot distinguish between the images due to underfitting maybe it would have worked better for more epochs(≥ 50)
4. Some model specially the Batch Normalised CNN gave loss= 1.62 which means it randomly guessing yet it gives better result than other models is a case of overfitting
5. We only judged the model on basis of accuracy but there can be more metrics required for specially medical purposes(important for the False-Negative Case) like the Precision, Recall (Sensitivity), Specificity, F1-Score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC) which will help in better decision making.

Conclusion

This experiment evaluates three distinct approaches to image classification: Custom Sequential CNNs, Deep Fixed-Architecture Networks (VGG), and Residual Learning (ResNet).

Custom CNNs (CNN1 & CNN2): These serve as the baseline, designed with 5 convolutional layers to establish a benchmark for feature extraction. CNN2 introduces Batch Normalization to stabilize the training process and accelerate convergence compared to the standard CNN1.

Deep Fixed Architectures (VGG16 & VGG19): These models significantly increase the network depth (16 and 19 layers, respectively). By using stacks of 3×3 convolutions, they capture more intricate hierarchical features. The study compares both a "from-scratch" implementation and a Transfer Learning approach with fine-tuning.

Residual Learning (ResNet50): Representing the most complex architecture in this study, ResNet50 utilizes Skip Connections to prevent the vanishing gradient problem. This allows the model to maintain high-performance levels even at a depth of 50 layers

Future work -:More metrics like **F1-score**, **AUC-ROC** can be used to better judge the models and **Explainable AI** techniques can be used to fine-tune the model as per more nuanced testing.

For more work on the project :<https://colab.research.google.com/drive/1TMBj1Ff3AUGWUb-t7PNSqAB2nRVHTXIJ?usp=sharing>

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